📘Clover SmartFit GUI – Final 2-Week Python Project Document

🧹Project Title

Clover SmartFit GUI: Personalized Size Recommendation System for Activewear and Scrubs

👜Objective

Build a complete Tkinter-based desktop GUI application that helps customers get personalized clothing size recommendations for products like biker shorts, scrubs, and sports bras. The system adjusts recommendations based on inputs like measurements, fit preferences, and activities.

🕒Timeline: 2 Weeks (Approx. 20 Hours)

Task	Est. Time
GUI Layout + Styling	4 hours
Input Handling & Validation	2 hours
Product-Specific Logic	5 hours
Recommendation System	3 hours
CSV Save / Reset / Exception Handling	2 hours
Testing & Polish	4 hours

🛠️Technologies Used

- **Python 3.10+**
- **Tkinter** (built-in GUI)
- **CSV Module** (for saving recommendations)

📂Project Structure

All in one Python file (`clover_smartfit.py`) for easy submission and testing.

```
clover_smartfit.py
├── GUI Layout (Tkinter widgets)
├── Input fields (bust, waist, hips, inseam)
├── Dropdowns (product type, activity, fit preferences)
├── Logic engine (conditional + boolean)
├── Recommendation output
├── Save to CSV
├── Reset form
└── Exception handling
```

📋Features & Functionalities

User Inputs

- Bust
- Waist
- Hips
- Inseam
- Product Type: Biker Shorts, Scrubs, Sports Bra
- Activity: Yoga, Running, Medical Work, etc.

🔦Dynamic Fields (based on Product Type)

- **Biker Shorts**:
- Length Preference (Mid-Thigh, Above-Knee, Below-Knee)
- **Scrubs**:
- Fit Preference (Slim, Loose, Baggy)
- Layer Check (Checkbox – size up if checked)
- **Sports Bra**:
- Padding Preference (Padded / Unpadded)

🔗Output

- Fit recommendation text
- Sizing suggestion
- Tips based on fit logic
- Optional save to CSV

👜Core Logic & Conditionals (Example)

```python
if product_type == "Scrubs":
    if fit_pref == "Slim":
```

```
        size_adj = -1
    elif fit_pref == "Baggy":
        size_adj = 1

    if wears_layers:
        size_adj += 1

    final_size = base_size + size_adj
```

```
if product_type == "Sports Bra" and padded and bust > 36:
    tip = "Padded options may offer more support for larger busts."
```

💾Sample Output (GUI)

```
✅ Recommendation:
Your best size is Medium.
We recommend a Loose fit for comfort.
Since you wear layers, we’ve sized you up for better movement.
```

💻Sample GUI Widgets Used

```
# Entry fields
bust_entry = tk.Entry(root)
waist_entry = tk.Entry(root)

# Dropdown for product type
product_var = tk.StringVar()
product_dropdown = ttk.Combobox(root, textvariable=product_var, values=["Biker Shorts", "Scrubs", "Sports Bra"])

# Checkbox for scrubs
layers_var = tk.BooleanVar()
tk.Checkbutton(root, text="Do you wear layers?", variable=layers_var)
```

📁Optional Extensions

- Connect to Order Form Generator
- Export all saved recommendations to CSV for admin use

🔗What You'll Demonstrate

- GUI development with Tkinter
- Variables, functions, conditionals, booleans
- Exception handling
- File I/O (CSV)
- User input management
- Real-world logic application in retail/clothing

🕤Testing Checklist

-

💴Final Submission

- One `.py` file
- Optional `.csv` file with saved recommendations
- Flowchart and algorithm (provided)

⛏️Simple Bullet-Point Algorithm

- Start the app
- Show the GUI with input fields
- User enters:
- Bust, Waist, Hips, Inseam
- Product Type (Biker Shorts, Scrubs, Sports Bra)
- Activity Type (e.g., Yoga, Running, Medical Work)
- Show more options depending on product:
- **If Biker Shorts** → Ask for length (Mid-Thigh, Above-Knee, Below-Knee)
- **If Scrubs** → Ask for fit (Slim, Loose, Baggy) and if they wear layers
- **If Sports Bra** → Ask if padded or unpadded
- Validate that all fields are filled correctly
- Apply product-specific logic:
- For Scrubs: adjust size based on fit + layer check
- For Sports Bra: give tip if bust is large and padded is selected
- For Biker Shorts: give tip based on length selected
- Create a final recommendation message
- Show result in the app
- Ask if user wants to:
- Save to CSV
- Reset form

- End